JOHN LI

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EDUCATION

Carnegie Mellon University — **Pittsburgh, PA**
Master's of Science in Mechanical Engineering Advanced Study — *Anticipated Graduation December 2027*
***Concentration*:** Design and Manufacturing of Mechanical Systems

Stony Brook University — **Stony Brook, NY**
Bachelor's Degree of Engineering in Mechanical Engineering — *2022- 2026*
***Honors*:** Dean's List (Top 20% of class) | University Scholar Program (Top 8% of the class) | GPA: 3.8

ENGINEERING PROJECT & LEADERSHIP

SBU Student Launch *Vehicle & Propulsion Subteam Lead | Technical Leadership & Team Management* — *June 2025 - Present*

- Managed full lifecycle development of two custom high-power competition airframes, engineering a Mars rover carrier and a precision altitude vehicle to achieve target apogees (1459 ft and 758 ft) within tight competition margins
- Conducted flight profile simulations using RockSim and MATLAB to optimize vehicle stability, verifying that aerodynamic center and center of gravity margins met structural constraints across the entire flight envelope
- Designed and fabricated robust single-deployment parachute recovery systems, utilizing electronic altimeters and black powder ejection charges to achieve controlled descent rates (12 ft/s and 19 ft/s) and ensure 100% vehicle reusability.
- Served as Technical Lead to direct a 5-member design sub-team and coordinated full system integration across 15+ payload and recovery members, managing manufacturing schedules, design reviews, and safety compliance from PDR through launch day

PROFESSIONAL EXPERIENCES

Stony Brook University Materials and Mechanics Lab — **Stony Brook, NY**
NASA Space Grant Intern & Undergraduate Research Assistant — *May 2025 - Present*

- Conducted ASTM D6484 open-hole compression (OHC) testing on woven FRP laminates (hole sizes 3–6 mm) using a Shimadzu UTM and high-speed imaging to characterize load response and fracture process zone (FPZ) damage.
- Analyzed sub-scale failure modes across varying geometries, finding first-damage gross stress converging at 42–46 MPa — isolating a load-introduction-limited failure mode and informing future test protocol improvements.
- Continuing to investigate contributing variables: specimen length, grip clamping conditions, and hole machining quality

Stony Brook University Division of Information Technology — **Stony Brook, NY**
Senior Lead IT Technician — *February 2023 - Present*

- Led and mentored a 25+ member technical team, delegating tasks effectively, and promoting a collaborative culture.
- Standardized workflows through SOPs and training programs, enhancing efficiency, team performance, and service quality.
- Directed cross-functional initiatives, aligning technical solutions with organizational goals and enhancing user satisfaction.

Brookhaven National Laboratory (BNL) — **Upton, NY**
Mechanical Engineering Intern — *June 2024 - August 2024*

- Utilized Creo and Ansys to model and analyze new structural testing components, ensuring all facility designs met strict engineering margins for the upcoming Electron-Ion Collider (EIC) accelerator program.
- Designed and engineered an innovative end cap and fixture for a critical 591MHz fundamental power coupler (FPC) superconducting radio frequency (SRF) component with guidance of the project manager.
- Prototyped an thermal test rig using additive manufacturing and embedded electronics, expanding facility capabilities to simulate heat transfer across multiple future bellows experiments
- Analyzed experimental thermal data to quantify contact resistance in RF-shielded bellows, directly validating component performance boundaries under simulated operational environments

TECHNICAL SKILLS

Softwares: Ansys, LabVIEW, MATLAB, Solidworks, Creo, Inventor, Fusion 360, Arduino IDE
Technical: Project Management, Design & Prototyping, Manufacturing Processes (3D Printing & Manual & CNC Machining)
Finite element Analysis, Circuit Building & Analysis, Microsoft Office, Google Suite